

Nandana K A

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SUMMARY

Software Development Engineer with 2+ years of experience building scalable backend systems and ML-driven pipelines for large-scale financial data processing. Proven track record of shipping production services that handle 30K+ bank statements per day, with measurable gains in matching accuracy, latency, and fraud detection. Comfortable across Python and R ecosystems, with hands-on experience in NLP, LLM integration (Vertex AI/Gemini), and cloud-native deployments.

EXPERIENCE

Software Development Engineer

August 2023 – Present

Perfios Software Solutions

Bangalore, India

- Designed and developed scalable backend systems for financial data processing, including fuzzy name matching and de-duplication, **improving match accuracy by 47%** and **reducing false positives by 25%**.
- Optimized the matching pipeline to **cut latency and memory usage by 50%**, enabling higher throughput on the same infrastructure.
- Built and maintained RESTful APIs in Flask with Celery for asynchronous task processing and PostgreSQL-backed schemas, reliably processing **30K+ bank statements per day**.
- Designed and deployed an on-prem backend API for fuzzy name matching using R Plumber, achieving **sub-200ms response times** through asynchronous execution with mirai, and enabling reproducible client deployments via Docker.
- Partnered with customers, sales, product, and QA to understand Malaysian naming patterns, and improved fuzzy matching by implementing a BiLSTM-based person/entity classifier with token alignment, achieving a **22% accuracy increase**.
- Built and optimized fraud detection engines analyzing large-scale bank transaction data using Polars and Pandas, integrating spend-category and transaction-mode intelligence for a **28% improvement in detection accuracy**.
- Developed a document Q&A chatbot on Vertex AI (Gemini) for insurance reports, enabling underwriters to ask natural-language questions and receive precise, context-driven answers through prompt engineering.

PUBLICATIONS

Human Activity Recognition for Office Surveillance | *IEEE INCET 2023*

May 2023

- Designed a deep-learning pipeline (ConvLSTM, GRCNN, LRCN) for activity recognition on real-world office surveillance video, covering preprocessing, feature extraction, and classification.
- Achieved **86.17% test accuracy** with LRCN, outperforming the ConvLSTM and GRCNN baselines on robustness and generalization.

TECHNICAL SKILLS

Languages: Python, R, SQL (PostgreSQL), Java, C/C++, HTML/CSS

Frameworks/Backend: Flask, Celery, R Plumber, RESTful APIs, FastAPI, Mirai

Data & AI: Polars, Pandas, NumPy, Vertex AI (Gemini), LLMs, Prompt Engineering, BiLSTM, NLP, Vector DB

Developer Tools: Docker, Git, Google Cloud Platform (GCP), VS Code, PyCharm, IntelliJ

EDUCATION

PES University

Bachelor of Technology in Computer Science; CGPA: 8.4/10.0

Bangalore, India

August 2019 – June 2023